

IE4-DBL: Lab 1 assignment

Deadline: Sunday Nov 2 - 11:59 PM for lab groups 1,2 and 3.

What to upload: One PDF file per team, with:

- your names
- your matriculation numbers
- solutions for assignments 1, 2 and 3

Where to upload? Moodle room, "Submission Lab 1"

In preparation of the lab, you must prepare the assignments and submit your solution in the Moodle room for this course. The submission must be a PDF file. During the lab, each team member must be able to present all parts of the solution.

You can use a drawing tool of your choice to create the diagrams or just draw the diagrams on a piece of paper. You can also use any DBMS to develop / test your SQL scripts.

If you have questions or issues, please ask them in the Moodle room.

Assignment 1: Student Information System

A university wants to store information about its students and their courses in a database. The following requirements have been identified:

- The university offers one or more study programs. Every study program has a unique program id, a unique name, and the required credit points to finish it.
- A study program is made up of one or more courses. A course belongs to exactly one study program and has a course id, a name, a description, and credit points. A course can have no, one or more other courses as a prerequisite.
- A student has a student id, first name, last name, date of birth and a year of enrollment. A student is enrolled in exactly one study program.
- A student attempts courses that are part of his/her study program. If a student attempts a course, the year and term (summer or winter semester) and grade (between 0 and 15 points) are recorded.

Assignments:

1. Create an entity-relationship-model according to these requirements, including entities, relationships and attributes. Mark the key attributes of each entity. Add for each relationship the cardinality and participation type.
2. Can you think of adding some additional composite, multivalued, or derived attributes in this example?
3. Convert the ERM into a relational schema.
4. Write SQL-statements that create the corresponding tables. Come up with reasonable constraints and datatypes for the fields of the tables.
5. Insert example data in the database you created (you could for example give rows for your own study program / courses / attempts or any fictional data).

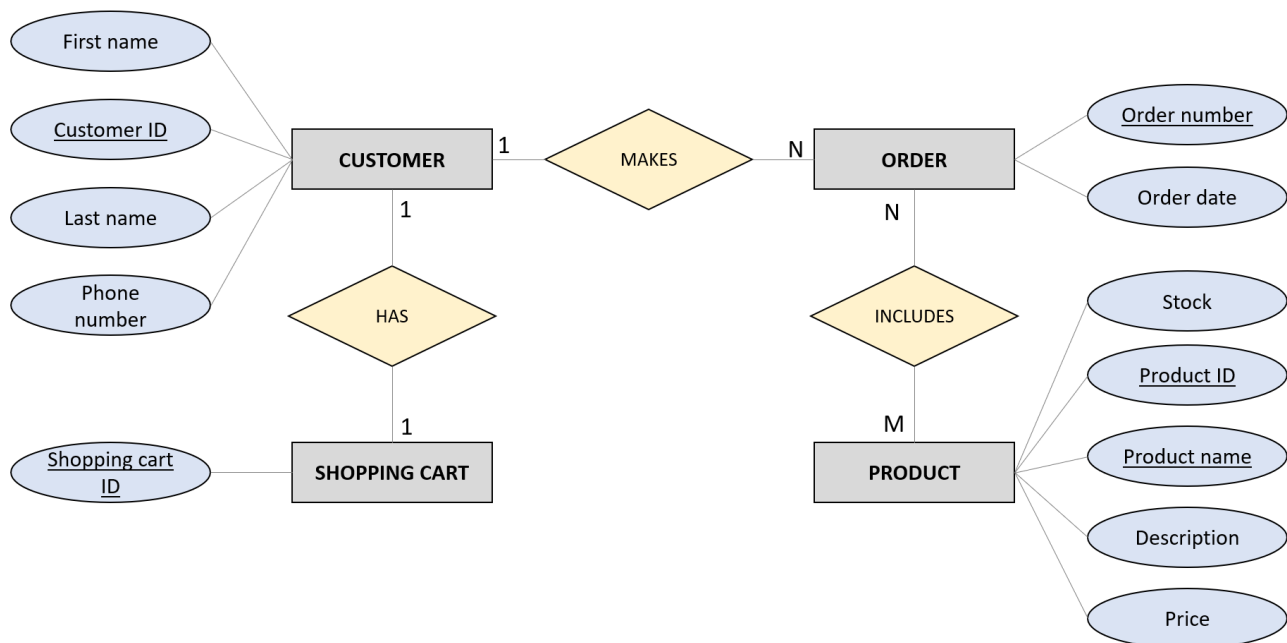
Assignment 2 ERD for a Technology Support Company

A technology company is developing a support ticket management system to handle customers' complaints efficiently and track the support staff's work. The system must store information about support requests, customers, support staff, and their interactions.

1. Create an entity-relationship-model according to these requirements, including entities, relationships and attributes. Mark the key attributes of each entity. Add for each relationship the cardinality and participation type.
2. Can you think of adding some additional composite, multivalued, or derived attributes in this example?
3. Convert the ERM into a relational schema.
4. Write SQL-statements that create the corresponding tables. Come up with reasonable constraints and datatypes for the fields of the tables.
5. Insert example data in the database you created.

Assignment 3: ERM for an Online Shopping Platform

The following entity relationship model is given. Assume that the database is filled with data according to the ERM. Decide whether the following statements are either true (T), false (F), or undecidable/maybe (U). "U" is used for statements that can be either T or F depending on the stored data. Evaluate the statements based exclusively on the ERM and the restrictions it contains.



No connection between s.c. and orders.

No	Statement	Answer
01	A customer can have multiple shopping carts. <i>customer - s.c. - 1:1</i>	F
02	A product can be included in multiple orders. <i>can be in only one or 0.</i>	U
03	Each shopping cart is associated with a specific order.	F
04	An order can contain only one product. <i>depends on the data.</i>	U
05	A customer can place multiple orders. <i>depends on the data.</i>	U
06	Each order has a unique order number.	T
07	Each order must be associated with a customer.	T
08	A customer can place an order without a shopping cart.	T
09	A product can be sold at different prices in different orders.	F
10	Every product can be contained several times in the same order.	F
11	A product can only be uniquely identified by the combination of the Product name and Product ID. <i>product ID is the primary key, not the same</i>	F

each product has one price.

makes and has relationships are separate.